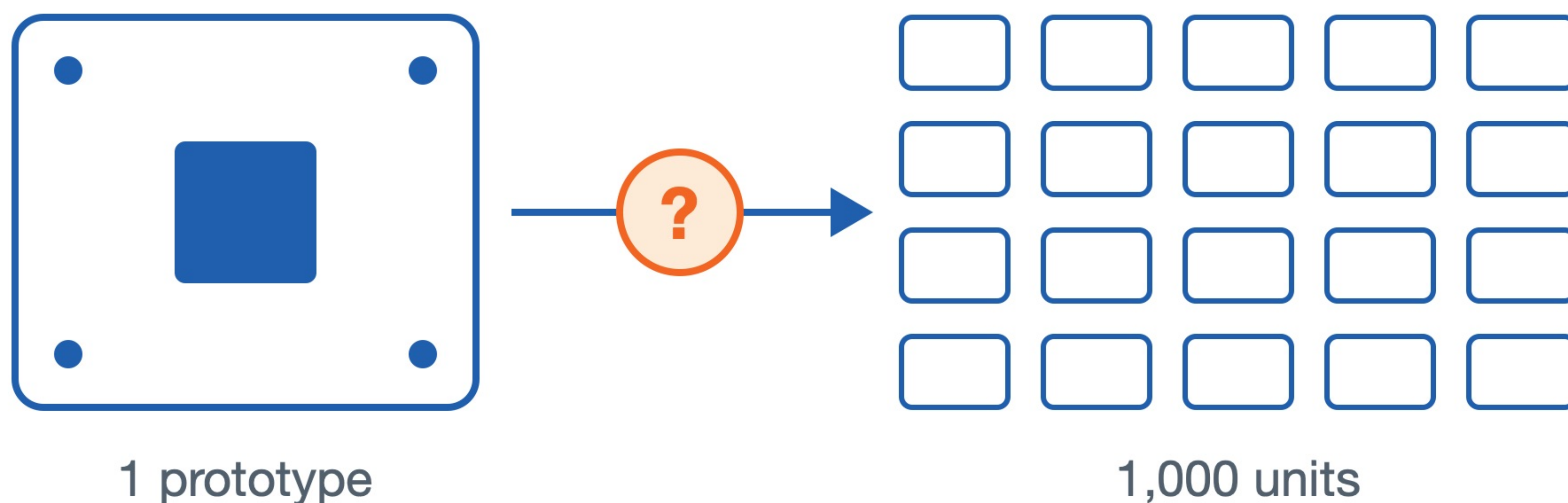


PROTOTYPE TO PRODUCTION

Works on the bench. Ready for 1,000?

Six things to lock before your
first production run.



Swipe for the checklist ->



TEST POINTS

No test pad, no test.

On the line, nobody probes a board by hand. A fixture presses pins onto test pads and runs a test firmware that checks power rails, radio and sensors in one pass.

Add the pads after layout freeze and you pay for a board respin.

LOCK

Test-pad list, fixture design, and the pass/fail limits for each test.

Every unit needs its own identity.

Firmware, serial number and per-device keys have to reach each board at the factory, without your engineers there. Then the debug port gets locked, or anyone with a probe can read it out.

For example: readout protection (RDP) on STM32, APPROTECT on Nordic nRF52.

LOCK

Where keys are generated, who holds them, and when debug is locked.

3

THE BOM

One missing part stops the line.

A prototype BOM is chosen from what is on the shelf. A production BOM has to survive lead times and end-of-life notices for as long as you sell the product.

Check every part's lifecycle status: Active, NRND (not recommended for new designs) or EOL.

LOCK

Lifecycle check, and a second source or alternate footprint for risky parts.

The bootloader ships once.

Application firmware can change for years. The bootloader on unit 1,000 usually stays as it shipped. Get it wrong, and the fix is a recall, not an update.

The flash partition layout is just as hard to change once devices are in the field.

LOCK

Partition layout, image signing, rollback, and the OTA path.



EMC PRE-COMPLIANCE

Scan before the lab does.

Formal EMC lab time is booked ahead.

Fail it after the PCB is frozen, and
the fix can mean a board respin.

An in-house pre-scan finds the noisy
trace while a fix is still cheap.

Typical targets: FCC Part 15 (US), IEC 60601-1-2
(medical EMC), IEC 61000-6 series (industrial).

LOCK

Target standards, pre-scan results,
and margin before booking the lab.



TRACEABILITY

Know which units share the fault.

When one unit fails in the field, the first question is: which others are the same? Link each serial number to its board revision, firmware version and component lots.

Medical device makers need this for ISO 13485 traceability anyway.

LOCK

Serial-number format, label, and the record written at end-of-line test.

TAKEAWAY

Lock it while it's on the bench.

- 1 Test pads and a production fixture
- 2 Programming, keys, locked debug
- 3 BOM lifecycle and second sources
- 4 Bootloader, partitions and OTA
- 5 EMC pre-scan before the lab
- 6 Serial-number traceability

None of these show up in a demo.
All of them show up in production.

Follow CoreFragment for more